

Project Report

Sensor Calibration and Time Drift Analysis of Urban Air Quality Sensors

Background:

Due to their affordability and ease of deployment, low- cost metal oxide gas sensors have become increasingly popular in urban air quality monitoring. However, these sensors have been known to suffer from cross sensitivity to co-pollutants, temperature, humidity, as well as performance degradation over time known as sensor drift. Without proper calibration, raw sensor output can deviate substantially from true pollutant concentrations.

Collected in Italy between March 2004 and April 2005, the UCI Air quality dataset provides a chance to study these occurrences. The dataset contains hourly reading from five metal oxide sensors along with ground truth measurements. The primary pollutant of interest, Carbon Monoxide (CO), has four other cross sensitivity sensors being non-methanic hydrocarbons (NMHC), nitrogen oxides (NO_x), nitrogen dioxide (NO₂), and ozone (O₃), as well as measurements for temperature, relative and absolute humidity.

Research Question:

Can machine learning models trained on raw sensor output along with environmental covariates accurately predict CO concentrations? And does the calibration relationship between sensors and ground truth remain stable over time, or is there evidence of sensor drift over time?

Data and Preparation:

The raw dataset contains 9,537 observations over 15 variables. Sentinel values of -200 used by the author for missing values, were replaced with NA. In order to arrange in chronological order, a DateTime variable was created to combine Date and Time columns. NMHC ground truth was rendered unusable as a response due to 90% missing values. CO ground truth and its respective sensor were only missing roughly 20% of observations, with no evidence of selection bias between usable and excluded rows.

A working dataset was built for modeling using the CO ground truth, all five sensors, temperature, relative humidity, and absolute humidity, then applying listwise deletion of missing values. This resulted in 5140 observations. Cyclic time features were built with sine and cosine transformations to preserve continuity across the midnight boundary. A deployment time index was created by hours since first deployment. Column names containing parentheses were renamed prior to modeling to avoid formula parsing issues in R. A chronological 70/30 train-test split was used, with 70% of the data for training and 30% for testing.

Methods and Results:

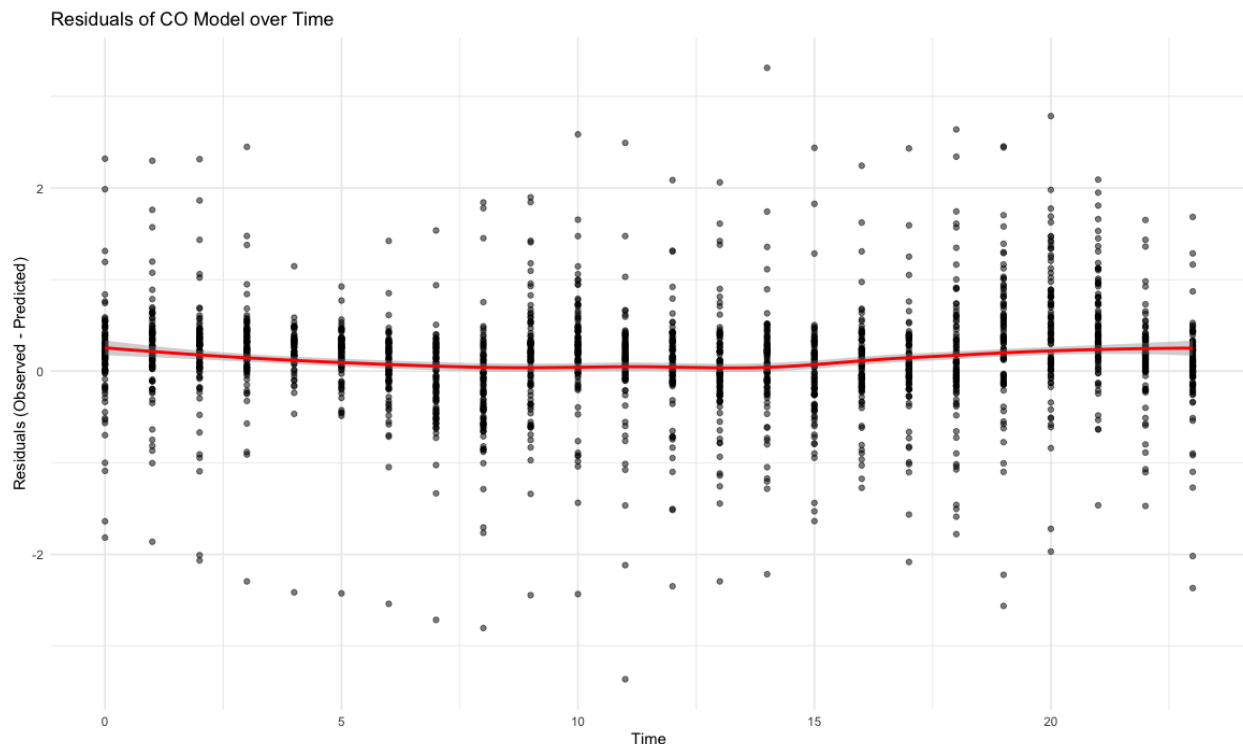
Multiple predictive models were fit and evaluated using root mean squared error. For a reference, a naive mean baseline was included. Models were compared both with and without lag features, motivated by autocorrelation diagnostics discussed below.

Block 1: Linear models and Regularization

Two ordinary least squares models were fit, one without temporal features and the other with cyclical hour encodings (hour_sin, hour_cos). Ridge and LASSO regression were also fit on scaled predictors using cross validated lambda selection. The scaling parameters were derived from the training set and applied to the test set to prevent data leakage. Among this group of the models, the linear model with time feature performed the best. Ridge and LASSO underperformed likely due to the cyclical features distorting their penalty term.

Block 2: PCA Regression and GAM

Principal component analysis was applied to the sensor and environmental variables. A linear model was fit on the first 5 principal components which together explain 97.9% of sensor variance. A generalized additive model (GAM) was also fit using penalized regression splines for each predictor and a cyclic cubic spline for the hour of day effect. The GAM was estimated using restricted maximum likelihood (REML) smoothing parameter selection. The GAM achieved an adjusted R-squared of 0.93.



Model	RMSE
Mean Baseline	1.414
Linear (no time)	0.601
Linear + time	0.599
Ridge	0.713
LASSO	0.821
PCA	0.648
GAM	0.626

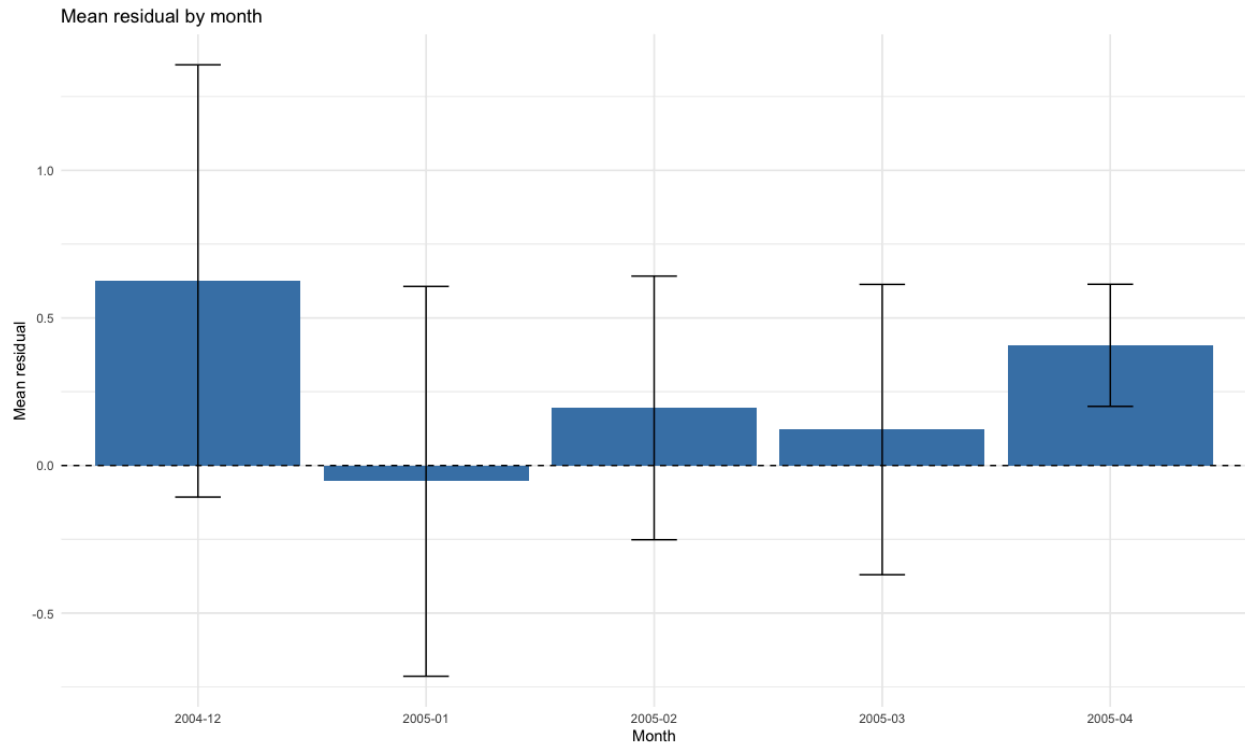
Table 1. Model RMSE comparison without lag features

Drift analysis:

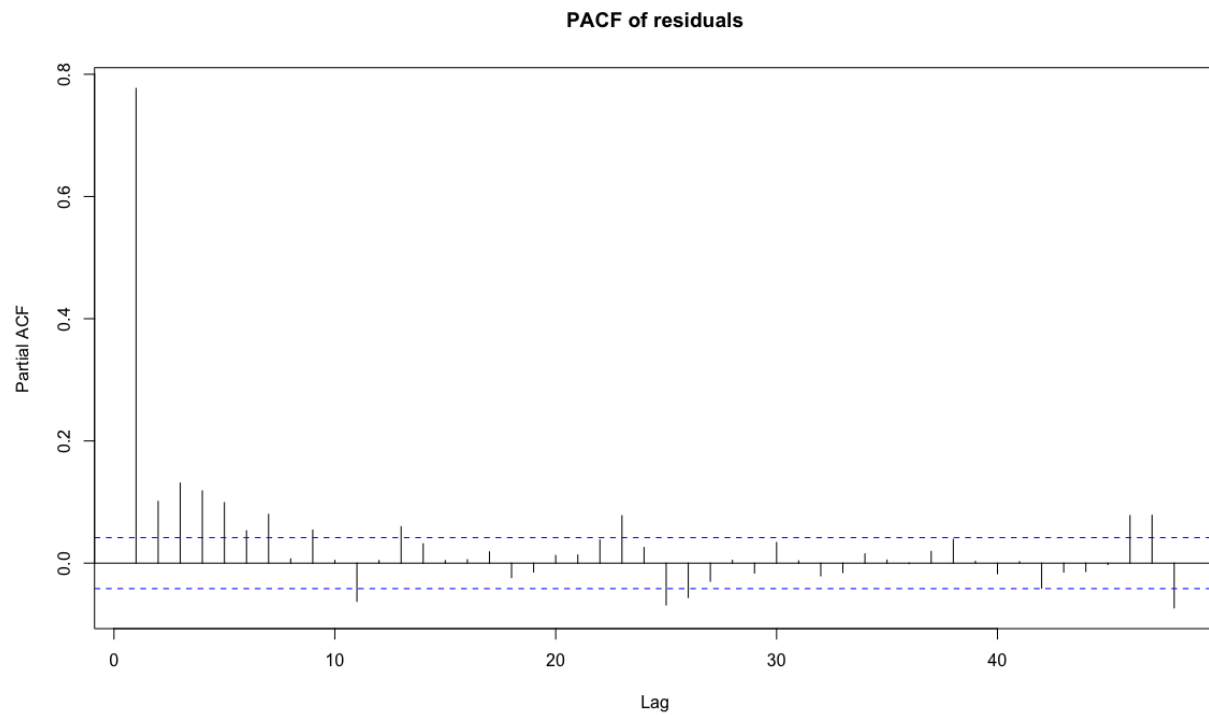
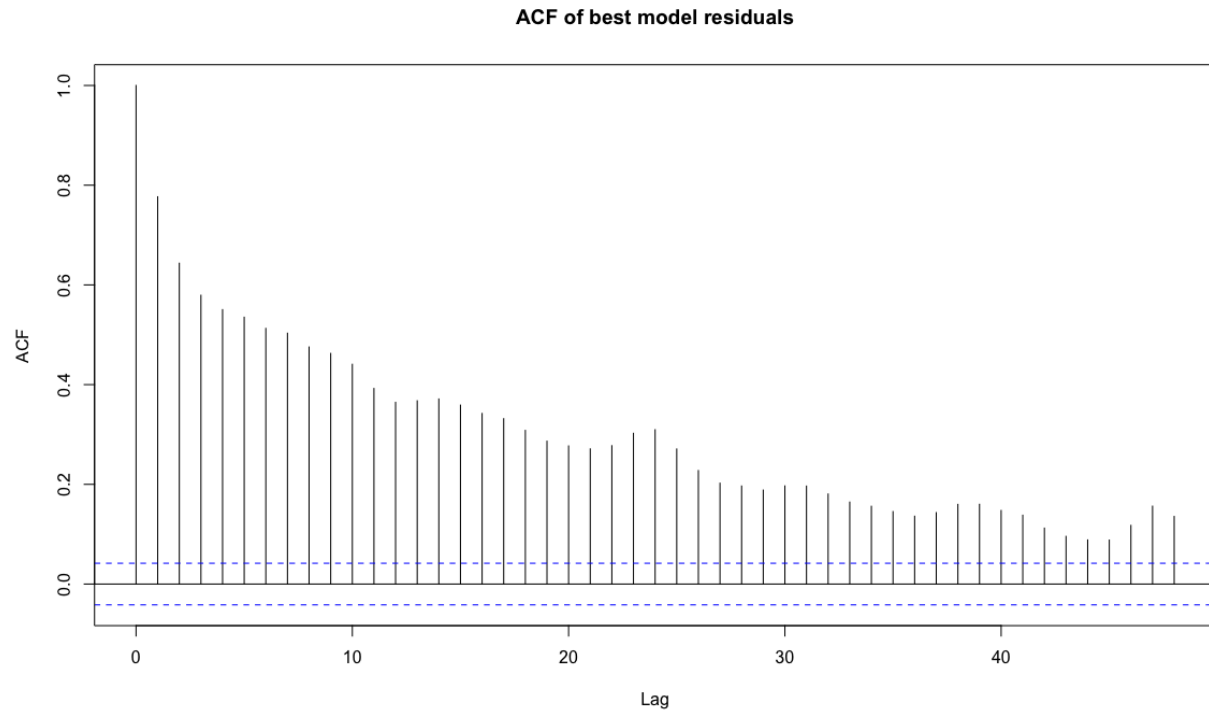
Following initial model fitting, the best performing model was used to generate test set residuals for temporal diagnostic analysis. Three complimentary methods were used to determine whether the sensor displayed drift over the deployment period.

First, a linear regression of residuals on the deployment time was first to test for a systematic trend in error over time. The estimated drift slope was $-1.5 \times 10^{-5} \text{ mg/m}^3$ per hour, which corresponds to -0.0111 mg/m^3 per month. There is no statistical significance with $p = 0.3682$, providing no evidence of sensor drift.

Second, the mean residuals were calculated by month. A distinct difference was observed between December (0.625) and January (-0.054), suggesting a seasonal bias more so than sensor degradation. Note that this pattern is consistent with differences in atmospheric conditions these winter months.

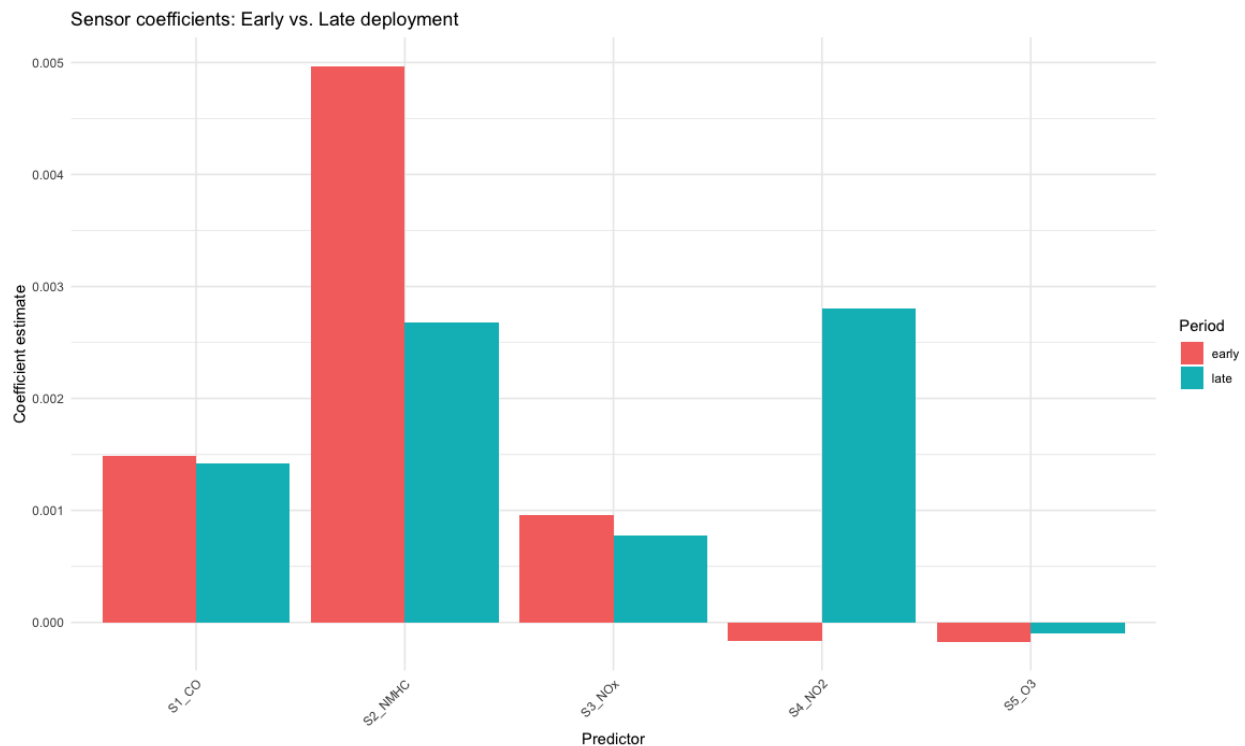


Third, an autocorrelation function (ACF) and partial autocorrelation function (PACF) were computed on the residuals. The ACF revealed strong and slowly decaying autocorrelation through lag 48, and the PACF showing a pronounced spike at lag 1 (approximately 0.78). This tells us that the model's errors were serially correlated. This is a known consequence of CO's physical persistence in the atmosphere and motivated the addition of lagged response features in subsequent modeling.



Finally, a direct coefficient stability test was performed by fitting separate linear models on the early and late halves of the dataset, then comparing sensor coefficients between periods. The CO sensor (S1_CO) changed by only 4.1% between early and late deployment, yielding

strong evidence that the sensor's calibration relationship is stable over time. The S2_NMHC cross sensitivity coefficient showed a larger shift (-46%), however, given its small magnitude, the impact on CO prediction is minimal. Instability in the temperature and humidity coefficients can be attributed to seasonal multicollinearity, rather than sensor degradation.



Block 3: Models with Lag Features:

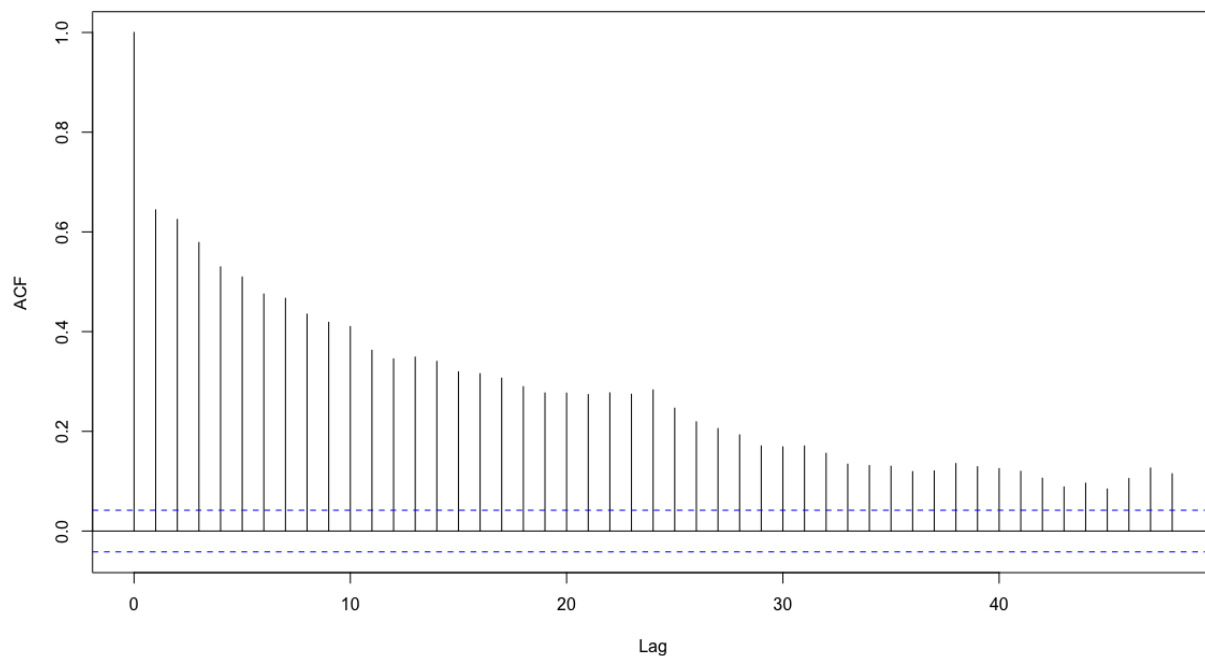
Motivated by the strong lag 1 autocorrelation identified by the ACF, lag features for the previous 3 hours of CO ground truth were added (CO_lag1, CO_lag2, CO_lag3). The lagged dataset was built from the full model dataset prior to splitting, then reapplying the chronological 70/30 split. All models were then refit identical to the prior procedures.

Every model improved upon the addition of lag features, confirming that the time persistence in CO concentration carries predictive information beyond what the sensor and environment provide alone. The best model in this instance was the linear model without explicit time features, achieving RMSE = 0.493, which is 18 % reduction from the lagless model. Notably, the cyclical hour encoding became largely redundant once the lag features were included.

Model	RMSE (no lag)	RMSE (with lag)	Improvement
Linear (no time)	0.601	0.493	18 %
Linear (with time)	0.599	0.494	17.5 %
Ridge	0.713	0.600	15.8 %
LASSO	0.821	0.735	10.5 %
PCA	0.648	0.647	0.2 %
GAM	0.626	0.516	17.6 %

Despite the improvement in RMSE, ACF on the lag model residuals shows that the autocorrelation was not resolved, thus lag features alone cannot address this.

ACF after adding lag1, lag2, lag3



Model Comparison:

The best overall model was the linear model with three lag features (RMSE = 0.493). The GAM with lag features performed similarly (RMSE = 0.516), closing the gap in the no lag comparison where GAM was penalized for overfitting. Ridge and LASSO consistently underperformed, attributed to penalty distortion caused by the cyclical time features and lag variables on a different scale. PCA showed the least benefit from the addition of lag features.

Being that a simple linear model outperformed more complex approaches is somewhat expected given the strong linear relationships seen in the exploratory data analysis.

Discussion:

The primary goal of sensor calibration was achieved. A linear model using raw sensor readings, environmental conditions, and three hours of lagged CO measurements yields a calibration function capable of substantially improving raw sensor output. An additional advantage is the model's interpretability: positive coefficients on S1_CO and S2_NMHC confirm both sensors contribute meaningfully to prediction, while the negative coefficient on temperature is consistent with known behavior of metal oxide sensors.

The drift analysis produced a clear and consistent set of findings. No statistically significant drift was detected in either the residual trend test or the coefficient stability analysis. The primary CO sensor coefficient was very stable across deployment. The seasonal bias observed is better interpreted as a model generalization limitation than hardware degradation, i.e. the training period covers spring through fall, where the test period covers winter.

The most significant remaining limitation of the calibration model is the persistent residual autocorrelation. While lag features meaningfully improved RMSE, they did not resolve the underlying time dependence. Therefore prediction intervals derived from the model would be too narrow. Also that the model's errors at any given hour are predictable from its recent hours, an exploitable structure that an ARIMA model could leverage.

Limitations:

Several limitations of this analysis should be noted. First, the modeling was restricted to CO as the response variable. Since the dataset contains ground truth for NO_x and NO₂, extending the calibration pipeline to these gases is an essential next step. Second, NMHC ground truth was 90 % missing, disabling its use as a response as well as limiting the ability to fully check its cross sensitivity effects. Third, the chronological train/test split means the train set is of a different season than the test set, making it difficult to differentiate between model generalization error and seasonal effects. Fourth, residual autocorrelation persisted after including lag features, indicating a dedicated time series model such as ARIMA would more fully explain the time structure of the data. Finally, ridge and LASSO underperformed compared to OLS, likely due to penalty on cyclical time and lag features; which could be addressed in future work.

References:

- UCI Machine Learning Repository, Air Quality Data Set
<https://archive.ics.uci.edu/dataset/360/air+quality>

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